Chapter 6 • Trigonometry

Solved Review Exercise 6

MCQs

1. The value of $\tan^{-1}2$ in radians is approximately:

a) $\frac{\pi}{2}$
b) $\frac{3\pi}{2}$
c) 1.11π
d) 1.11

Ans: d

$\tan^{-1}2 = 1.107...$ rad.

2. In a right triangle, hypotenuse 13 and an angle 30° give opposite side:

a) 6.5 units
b) 7.5 units
c) 6 units
d) 5 units

Ans: a

Opposite $= 13\sin30° = 6.5$.

3. From 50 m away, a building's top has elevation 45°. Its height is:

a) 50 m
b) 25 m
c) 35 m
d) 70 m

Ans: a

$h = 50\tan45° = 50$ m.

4. $\sec^2\theta - \tan^2\theta =$

a) $\sin^2\theta$
b) 1
c) $\cos^2\theta$
d) $\cot^2\theta$

Ans: b

This is the Pythagorean identity.

5. If $\sin\theta = \frac{3}{5}$ and θ is acute, $\cos^2\theta =$

a) $\frac{7}{25}$
b) $\frac{24}{25}$
c) $\frac{16}{25}$
d) $\frac{4}{25}$

Ans: c

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$

6. $\frac{5\pi}{24}$ rad equals:

a) 30°
b) 37.5°
c) 45°
d) 52.5°

Ans: b

Multiply by $\frac{180}{\pi}$

7. 292.5° equals:

a) $\frac{17\pi}{6}$
b) $\frac{17\pi}{4}$
c) 1.6π
d) 1.625π

Ans: d

$\frac{292.5}{180} = 1.625$.

8. Which is a valid identity?

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a) $\cos(\frac{\pi}{2}-\theta)=\sin\theta$

b) $\cos(\frac{\pi}{2}-\theta)=\cos\theta$

c) $\cos(\frac{\pi}{2}-\theta)=\sec\theta$

d) $\cos(\frac{\pi}{2}-\theta)=\operatorname{cosec}\theta$

Ans: a

Cosine of the complement equals sine.

9. sin60°=

a) 1

b) $\frac{1}{2}$

c) $\sqrt{(3^2)}$

d) $\frac{\sqrt{3}}{2}$

Ans: d

The standard exact value is $\frac{\sqrt{3}}{2}$

10. $\cos^2 100\pi+\sin^2 100\pi=$

a) 1

b) 2

c) 3

d) 4

Ans: a

$\sin^2 x+\cos^2 x=1$ for every x.

2(a). Convert degrees to radians in terms of π.

(i) $255°\times\frac{\pi}{180}=\frac{17\pi}{12}$

(ii) $75°45'=75.75°=\frac{303}{4}$ degrees; multiply by $\frac{\pi}{180}$ to get $\frac{101\pi}{240}$

(iii) $142.5°\times\frac{\pi}{180}=\frac{19\pi}{24}$

2(b). Convert radians to degrees and minutes.

(i) $\frac{17\pi}{24}\times\frac{180}{\pi}$ = 127.5°
= 127°30′.

(ii) $\frac{7\pi}{12}\times\frac{180}{\pi}=105°$.

(iii) $\frac{11\pi}{16}\times\frac{180}{\pi}$ = 123.75°
= 123°45′.

3(i). Prove $\frac{\sin\theta}{1-\cos\theta}=\frac{1+\cos\theta}{\sin\theta}$

$\frac{\sin\theta(1+\cos\theta)}{1-\cos^2\theta}=\frac{1+\cos\theta}{\sin\theta}$

3(ii). Prove $\sin\theta(\operatorname{cosec}\theta-\sin\theta)=\frac{1}{\sec^2\theta}$

Left = $1-\sin^2\theta$
= $\cos^2\theta$
= $\frac{1}{\sec^2\theta}$

3(iii). Prove $\frac{\operatorname{cosec}\theta-\sec\theta}{\operatorname{cosec}\theta+\sec\theta}=\frac{1-\tan\theta}{1+\tan\theta}$.

Left=$\frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}$. Right, after multiplying by cosθ, is identical.

3(iv). Prove $\tan\theta+\cot\theta=\frac{1}{\sin\theta\cos\theta}$.

Left = $\frac{\sin\theta}{\cos\theta}\frac{+\cos\theta}{\sin\theta}$
= $\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}$
= $\frac{1}{\sin\theta\cos\theta}$.

3(v). Prove $\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\frac{+(\cos\theta-\sin\theta)}{\cos\theta+\sin\theta}=\frac{2}{1-2\sin^2\theta}$.

Combined numerator = $(\cos\theta+\sin\theta)^2+(\cos\theta-\sin\theta)^2$

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= 2; denominator
= $\cos^2\theta - \sin^2\theta$
= $1 - 2\sin^2\theta$.

3(vi). Prove $\frac{1+\cos\theta}{1-\cos\theta} = (\text{cosec}\theta + \cot\theta)^2$.

Right = $[\frac{1+\cos\theta}{\sin\theta}]^2$
= $(1+\cos\theta)^2 \frac{}{1-\cos^2\theta}$
= $\frac{1+\cos\theta}{1-\cos\theta}$.

4. If $\tan\theta = \frac{3}{\sqrt{2}}$ in Quadrant I, find the remaining ratios.

Use opposite=3, adjacent=√2, hypotenuse=√(9+2)=√11.
$\sin\theta = \frac{3}{\sqrt{11}}$; $\cos\theta = \sqrt{(\frac{2}{11})}$; $\text{cosec}\theta = \frac{\sqrt{11}}{3}$; $\sec\theta = \sqrt{(\frac{11}{2})}$; $\cot\theta = \frac{\sqrt{2}}{3}$

5. The top of a 30 m building has elevation 28°. Find the distance from its base.

d = $\frac{30}{\tan 28°}$
= $30\cot 28°$.
56.42 m approximately.

6. A 10 m ladder makes 65° with the ground. Find the height reached.

h=10sin65°.
9.06 m approximately.

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